



WATERSHED SEGMENTATION

STUDY NOTES

Luke Camilleri, Jake Desira, Kurt Abela, Tristin Bezzina, Nathan Vella

ARI2129



Contents

1. Introduction	2
2. Topographical Metaphor	2
2.1 From Image to Landscape	2
2.2 Key Definitions	3
2.3 The Flooding Simulation	3
2.4 Beucher's Contribution and Historical Context	4
3. Gradient-Based Watershed	4
3.1 Motivation: Why Not Apply Watershed Directly to Intensity?	4
3.2 The Morphological Gradient	4
3.3 The Flooding Algorithm on $g(f)$	5
3.4 Worked Numerical Example	5
3.4.1: Compute the morphological gradient	6
3.4.2: Sort and flood, pixels processed in ascending order of $g(p_i)$:	6
3.4.3: Results	6
3.5 Application: The Bubble Segmentation Example	7
4. Markers and Over Segmentation	7
4.1 The Over-Segmentation Problem	7
4.2 What Is a Marker?	7
4.3 Marker Imposition	7
4.4 Five-Step Procedure	8
4.5 Practical Effect	8
5. Failure Modes	8
Over-segmentation	8
Boundary leakage and under-segmentation	9
Noise and artifact sensitivity.	9
6. Alternatives to Watershed Segmentation	9
7. Real-World Applications	10
Quick reference	10
References:	12

1. Introduction

Image segmentation is one of the most fundamental problems in computer vision. Given a digital image, partition it into meaningful regions corresponding to objects or structures of interest. Applications range from delineating tumour boundaries in medical scans to separating cells in histology and identifying landcover in satellite imagery.

Among classical approaches, the watershed transform occupies a distinctive position. Rather than treating the image as a flat signal, it treats pixel intensities as elevations on a three-dimensional landscape. Water falling on this terrain flows downhill into valleys, forming lakes. Where floodwaters from two adjacent lakes would meet, a dam is built. Those dams trace the boundaries between image regions. This intuition was formalised by Beucher and colleagues at the Centre de Morphologie Mathématique (CMM), Mines Paris, establishing the vocabulary, catchment basins, watershed lines, regional minima, that remains standard today.

These notes cover the topographical metaphor and mathematical foundations (Section 2), gradient-based watershed with a worked numerical example (Section 3), markers and over-segmentation (Section 4), failure modes (Section 5), alternatives (Section 6), and real-world applications (Section 7), followed by a Quick Reference cheat sheet and curated Key Papers.

2. Topographical Metaphor

2.1 From Image to Landscape

The central insight of watershed segmentation is a change of representation: a greyscale image is not merely a two-dimensional array of intensity values, but a topographic surface embedded in three-dimensional space. Formally, let $f : D \subseteq \mathbb{Z}^2 \rightarrow \mathbb{Z}$ be a digital greyscale image defined on a finite rectangular domain D , where $f(x, y)$ denotes the intensity of the pixel at position (x, y) . The topographic surface S^p associated with f is the graph:

$$S^p = \{ (x, y, z) \in \mathbb{Z}^2 * \mathbb{Z} : z = f(x, y) \}$$

embedded in the three-dimensional lattice $\mathbb{Z}^3$. Under this mapping, each pixel (x, y) corresponds to a point on a surface at height $f(x, y)$. Bright pixels, those with high intensity, correspond to elevated terrain; dark pixels correspond to valleys or depressions. This reinterpretation turns the problem of segmenting an image into the problem of partitioning a terrain into drainage regions, a question studied in geomorphology long before its application to image analysis.

2.2 Key Definitions

A regional minimum of f is a connected set $M \subseteq D$ where $f(p) = c$ for all $p \in M$, and $f(q) > c$ for every adjacent pixel q outside M . In topographic terms, it is a valley floor from which any step outward moves to higher ground.

The catchment basin $CB(m_i)$ of a regional minimum m_i is the set of all pixels $p \in D$ such that a drop of water placed at p and following the path of steepest descent would eventually reach m_i . Formally,

$$CB(m_i) = \{ p \in D : \text{the steepest-descent path from } p \text{ terminates at } m_i \}$$

The catchment basins partition D into disjoint regions, each associated with one regional minimum, typically corresponding to coherent areas of uniform intensity separated by edges. The watershed line (or watershed transform) $W(f)$ is the complement of the union of all catchment basins:

$$W(f) = D \setminus \bigcup_i CB(m_i)$$

Geometrically, $W(f)$ consists of pixels on ridges of S^p , points from which water could flow to two or more distinct minima, and forms a closed connected boundary set dividing D into as many regions as there are regional minima.

2.3 The Flooding Simulation

Beucher formalised the watershed transform using the metaphor of progressive flooding from below. Consider the topographic surface S^p submerged upside-down in a body of water, or equivalently, imagine water rising uniformly from the lowest points of the landscape upward. At a given water level h , the flooded region is

$$T_h(f) = \{ p \in D : f(p) \leq h \}$$

As h increases from the global minimum to the global maximum of f , water first floods the deepest regional minima, forming isolated lakes. As the level rises, each lake expands outward. The critical event occurs when two lakes from different regional minima are about to merge: at precisely this level, a dam is constructed along the boundary between them. This dam, a single-pixel-wide ridge, is added to the watershed line. The flooding continues, with dams preventing the merging of basins, until the entire surface is submerged.

The flooding simulation is not merely a visualisation device. It provides a constructive definition of the watershed transform and directly motivates efficient algorithmic implementations.

2.4 Beucher's Contribution and Historical Context

The watershed concept was pioneered by Beucher and Lantuéjoul at the CMM in the late 1970s within the mathematical morphology framework of Matheron and Serra. Beucher's key contribution was establishing that the topographic metaphor is a mathematically precise correspondence, not merely an analogy. This had two consequences: first, properties such as the thinness of the watershed line and completeness of the basin partition could be formally proved rather than observed empirically; second, it suggested a natural generalisation, any scalar function on a discrete domain can be treated as a topographic surface, including crucially the gradient magnitude of a greyscale image, which forms the basis for the gradient-based watershed in Section 3.

3. Gradient-Based Watershed

3.1 Motivation: Why Not Apply Watershed Directly to Intensity?

A natural first attempt would be to apply the watershed transform directly to the greyscale image f . Under this approach the regional minima of f , the darkest points in each region, would seed catchment basins, and watershed lines would trace boundaries between them. In practice this fails for most natural images. Objects are defined not by their intensity minima but by the transitions between regions. A dark cell against a bright background has its lowest intensity values deep within its interior, far from the boundary we want to detect. Watershed applied to f places basin seeds inside the cell rather than at its edges, producing lines with no useful relationship to object boundaries.

The solution, formalised by Beucher and Meyer, is to apply the watershed transform to the gradient magnitude $g(f)$ instead. The gradient is high wherever the image changes rapidly, precisely at transitions between objects, and low wherever the image is smooth and homogeneous. The regional minima of $g(f)$ therefore correspond to object interiors, and the watershed lines of $g(f)$ correspond to object boundaries. This reframing transform watershed from a method that detects intensity minima into one that detects edges.

3.2 The Morphological Gradient

While several gradient operators can be used, including the Sobel, Prewitt, and Canny edge detectors, the morphological gradient is particularly well suited to watershed segmentation because it is defined within the same mathematical morphology framework as the transform itself. As defined by Beucher and Meyer, the morphological gradient of f with respect to a structuring element B is:

$$g(f) = (f \ [+] \ B) - (f \ [-] \ B)$$

where $[+]$ and $[-]$ denote the morphological dilation and erosion operators respectively. For a pixel at position x , these are defined as:

$$(f \oplus B)(x) = \max\{f(y) : y \in Bx\}$$

$$(f \ominus B)(x) = \min\{f(y) : y \in Bx\}$$

where $Bx = \{x + b : b \in B\}$ is the structuring element B translated to position x . The dilation at x takes the maximum intensity in a local neighbourhood defined by B , and the erosion takes the minimum. The morphological gradient is their difference: a measure of the spread of intensity values within that neighbourhood.

The structuring element B is typically a small symmetric shape: a 3x3 square or cross in 2D, or a symmetric interval of length 3 in 1D. The choice of B controls the spatial scale at which intensity transitions are detected. A larger B will smooth over fine details and emphasise coarser boundaries, while a smaller B responds to sharper, finer-scale transitions.

where $Bx = \{x + b : b \in B\}$ is the structuring element B translated to position x . The dilation takes the maximum intensity in the local neighbourhood; the erosion takes the minimum. Their difference $g(f)$ equals zero on flat regions and is large wherever intensity changes rapidly, making it a natural surface for watershed, with regional minima inside object interiors and ridges at object boundaries.

3.3 The Flooding Algorithm on $g(f)$

With $g(f)$ computed, the gradient-based watershed proceeds by applying the immersion flooding process to the gradient surface. In practice, the standard implementation uses a priority queue: all pixels are sorted in ascending order of their gradient value and processed from lowest to highest, with basin labels propagating outward level by level.

For each pixel dequeued, its labelled neighbours are examined. If all belong to the same basin, the pixel joins that basin. If neighbours from two or more distinct basins are found, the pixel is marked WSHED. If no labelled neighbours exist, the pixel seeds a new basin and propagates its label to unlabelled neighbours at the same gradient level via a FIFO sub-queue. This continues until all pixels are labelled or marked WSHED.

3.4 Worked Numerical Example

Consider the following one-dimensional signal of seven pixels p_1 through p_7 , representing a cross-section through an image with two dark objects separated by a bright peak:

Pixel	p_1	p_2	p_3	p_4	p_5	p_6	p_7
$f(p_i)$	3	1	2	8	2	1	3

The signal has two intensity minima ($p_2 = 1$ and $p_6 = 1$), representing the interiors of two dark objects, and a central peak ($p_4 = 8$), representing a bright boundary region between them.

3.4.1: Compute the morphological gradient

using $B = \{-1, 0, +1\}$ with mirror-padding at the boundaries:

$$g(p_i) = \max\{f(p_{i-1}), f(p_i), f(p_{i+1})\} - \min\{f(p_{i-1}), f(p_i), f(p_{i+1})\}$$

Pixel	Neighbours	Dilation	Erosion	$g(p_i)$	Regional min?
p1	3, 3, 1	3	1	2	Yes
p2	3, 1, 2	3	1	2	Yes
p3	1, 2, 8	8	1	7	-
p4	2, 8, 2	8	2	6	Yes
p5	8, 2, 1	8	1	7	-
p6	2, 1, 3	3	1	2	Yes
p7	1, 3, 3	3	1	2	Yes

The resulting gradient signal is $g = [2, 2, 7, 6, 7, 2, 2]$. There are three regional minima: $\{p1, p2\}$ at $g = 2$ (left dark object), $\{p4\}$ at $g = 6$ (bright peak, locally flat at its apex), and $\{p6, p7\}$ at $g = 2$ (right dark object). Note that p4, despite being the bright intensity peak in f , is a regional minimum in the gradient signal g because its gradient value of 6 is lower than both its neighbours ($g(p3) = g(p5) = 7$). This reflects the fact that the gradient measures local spread, not intensity height, the apex of a smooth peak has less local variation than the steep slopes flanking it.

3.4.2: Sort and flood, pixels processed in ascending order of $g(p_i)$:

Step	h	Pixel	Decision	State [p1...p7]
1	2	p1	No labelled neighbours → seed Basin B1	[B1, --, --, --, --, --, --]
2	2	p2	Neighbour p1 = B1 → assign B1	[B1, B1, --, --, --, --, --]
3	2	p6	No labelled neighbours → seed Basin B2	[B1, B1, --, --, --, B2, --]
4	2	p7	Neighbour p6 = B2 → assign B2	[B1, B1, --, --, --, B2, B2]
5	6	p4	No labelled neighbours → seed Basin B3	[B1, B1, --, B3, --, B2, B2]
6	7	p3	Neighbours p2=B1, p4=B3 → two basins → WSHED	[B1, B1, W, B3, --, B2, B2]
7	7	p5	Neighbours p4=B3, p6=B2 → two basins → WSHED	[B1, B1, W, B3, W, B2, B2]

3.4.3: Results

The final segmentation gives Basin B1 = $\{p1, p2\}$, Basin B3 = $\{p4\}$, Basin B2 = $\{p6, p7\}$, with watershed lines at $W = \{p3, p5\}$. The watershed lines fall precisely at the steepest transitions in the original signal, confirming that gradient-based watershed correctly localises boundaries at intensity edges.

3.5 Application: The Bubble Segmentation Example

Beucher and Meyer demonstrate the failure of direct intensity watershed using a greyscale image of circular bubbles. Applying watershed to f produces poor results because the regional minima lie deep within bubble interiors. Applying the morphological gradient $g(f)$ first and then running watershed correctly traces the bubble contours, since the gradient is high along the bubble walls and low inside. This confirms the core practical lesson: the method succeeds when object boundaries correspond to sharp local intensity transitions.

4. Markers and Over Segmentation

4.1 The Over-Segmentation Problem

The gradient-based watershed transform is theoretically complete but practically problematic: it treats every local minimum of the gradient surface as the seed for a distinct catchment basin. Real images contain hundreds or thousands of spurious minima caused by noise, texture variation, and quantisation artefacts. The result is over-segmentation, a partition containing far more regions than meaningful objects.

Formally, if gradient image g has k regional minima, the watershed transform produces exactly k catchment basins. For a typical 256×256 greyscale image, k may exceed 10,000, rendering the output analytically useless without further processing.

4.2 What Is a Marker?

A marker is a connected set of pixels used to identify a region of interest before flooding begins. Two classes exist. Internal markers are blobs of pixels located inside each foreground object, one per object to be segmented, seeding that object's catchment basin. External markers are pixels known to belong to the background, forming a single additional basin that absorbs all unmarked space. Markers may be placed manually by an expert or computed automatically using morphological operations such as opening-by-reconstruction and closing-by-reconstruction (Kaur & Jindal, 2011).

4.3 Marker Imposition

Once foreground markers M_{fg} and background markers M_{bg} are identified, the gradient image g is modified so that its only regional minima coincide with those marker locations. This is achieved through geodesic reconstruction, a morphological fill that converts all unmarked minima into non-minimum plateaux, preventing them from generating new catchment basins. The modified gradient g^* satisfies:

$$\text{Regional_Minima}(g^*) = M_{fg} \cup M_{bg}$$

Watershed is then applied to g^* , producing exactly $|M_{fg}| + 1$ catchment basins: one per foreground object plus one background region.

4.4 Five-Step Procedure

The standard marker-controlled pipeline proceeds as follows (Kaur & Jindal, 2011):

1. Compute the segmentation function (typically the morphological gradient g).
2. Compute foreground markers connected blobs inside each foreground object.
3. Compute background markers, pixels confirmed not to belong to any object.
4. Impose markers on g via geodesic reconstruction to yield modified gradient g^* .
5. Apply the watershed transform to g^* .

4.5 Practical Effect

On a benchmark lung CT slice, Kaur and Jindal (2011) show that plain watershed produces a heavily fragmented partition, while marker-controlled watershed reduces this to anatomically meaningful structures, cleanly delineating the lung boundary with no spurious interior splits. The key insight is simple: the number of output regions is controlled by the analyst, not by the noise floor of the image.

5. Failure Modes

Despite its theoretical rigour, gradient-based watershed exhibits well-documented failure modes that limit its reliability in practice. Gilani et al. (2025) conduct an empirical reliability audit of marker-free watershed segmentation on dermoscopic images, reporting catastrophic failure rates, defined as cases where the Dice overlap score falls below 0.1, of between 20% and 33% even under clean imaging conditions, a figure that worsens substantially under noise, blur, and contrast degradation.

Over-segmentation.

The most cited failure mode arises when the gradient image $g(f)$ contains too many regional minima. Noise, fine texture, and minor intensity fluctuations each introduce spurious local minima, and each minimum seeds its own catchment basin. The result is a fragmented partition containing many small, meaningless regions where only a few objects exist. Bai et al. (2017) identify this as the central limitation of classical watershed and the primary motivation for replacing the hand-crafted gradient with a learned energy surface.

Boundary leakage and under-segmentation.

Where the gradient is weak, at low-contrast boundaries where two regions transition gradually, the ridge between two potential basins is too shallow to form a stable watershed line. The flooding process merges basins across the intended boundary, producing a single region where two distinct objects exist. Gilani et al. identify this as the most clinically dangerous failure mode, since a missed boundary in a medical context corresponds directly to a missed lesion.

Noise and artifact sensitivity.

Gradient operators amplify high-frequency noise: small intensity perturbations produce large local gradient values, introducing spurious minima throughout $g(f)$. Gilani et al. demonstrate that under controlled noise and blur perturbations, watershed catastrophic failure rates increase to levels that make the method unreliable without dedicated preprocessing or marker-based constraints.

6. Alternatives to Watershed Segmentation

When watershed's gradient-dependent flooding produces unreliable results, two principal alternatives offer distinct approaches. Graph Cuts reframe segmentation as a global energy minimisation problem. Peng and Liu (2010) define the energy as:

$$E = \sum V_{\{p,q\}} + \sum D_p$$

where $V_{\{p,q\}}$ penalises discontinuities between neighbouring pixels and D_p measures how well a label fits observed intensity. The global minimum is found via max flow/min cut, unlike watershed's greedy local flooding. Peng and Liu (2010) show that using watershed to pre-segment the image into regions before applying graph cut reduces running time from 2.44 to 0.63 seconds on a 256×256 image, while improving boundary accuracy over either method alone.

Active Contours initialise a curve near an object boundary and iteratively deform it to minimise an energy functional. Baswaraj et al. (2012) identify a key advantage: active contours always produce continuous boundaries, whereas gradient-based methods such as watershed frequently produce discontinuous ones. Two variants exist, snakes, which deform an explicit set of contour points, and level sets, which represent the contour implicitly and can handle topology changes such as splitting and merging. Baswaraj et al. further distinguish edge-based active contours, which inherit watershed's sensitivity to noise, from region-based ones, which use global intensity statistics and require no specific initialisation.

Both alternatives are computationally heavier than watershed's $O(n)$ flooding. Watershed remains preferable when no labelled seeds are available and images contain strong, well-defined gradient ridges.

7. Real-World Applications

Watershed segmentation appears across medical imaging, industrial inspection, and computer vision research. Kaur and Jindal (2011) demonstrate marker-controlled watershed for lung lobe segmentation from CT data. Gilani et al. (2025) apply it to dermoscopic lesion boundary detection, establishing reliability benchmarks for clinical deployment. Niu et al. (2025) deploy an adaptive watershed framework for real-time particle size measurement in dynamic fluid flows. Bai and Urtasun (2017) use watershed as the backbone of an instance segmentation system for autonomous driving, replacing the hand-crafted gradient with a learned energy map. In each case the method succeeds where object boundaries correspond to high-gradient transitions.

Quick reference

Core formula: morphological gradient: $g(f) = (f \oplus B) - (f \ominus B)$

Flooding decision rule (per pixel p removed from queue):

- All labelled neighbours same basin $\rightarrow$ assign p to that basin
- Labelled neighbours from ≥ 2 basins $\rightarrow$ mark p as WSHED
- No labelled neighbours $\rightarrow$ seed new basin, assign new label

Function	Description
$(f \oplus B)(x) = \max\{f(y) : y \in Bx\}$	Dilation takes maximum intensity in local neighbourhood
$(f \ominus B)(x) = \min\{f(y) : y \in Bx\}$	Erosion takes minimum intensity in local neighbourhood
$T_h(f) = \{p \in D : f(p) \leq h\}$	Flooded region at water level h
$W(f) = D \setminus \bigcup_i CB(m_i)$	Watershed line as complement of all catchment basins
$Regional_Minima(g^*) = M_{fg} \cup M_{bg}$	Condition satisfied after marker imposition

Parameter	Guidance
Structuring element B	Small symmetric shape, 3×3 square or cross in 2D, interval of 3 in 1D
Water level h	Ranges from global minimum to global maximum of f

<i>Number of markers</i>	One internal marker per object; one external marker for background
<i>Catastrophic failure threshold</i>	Dice score < 0.1 (Gilani et al., 2025)

Misconception	Correction
<i>Watershed understands objects</i>	It only follows gradient structure, it has no concept of what an object is
<i>More regions means better segmentation</i>	Too many regions indicates over-segmentation, not precision
<i>Distance transform is the same as watershed</i>	Distance transform is used to compute markers before watershed is applied
<i>A bright intensity peak is always a boundary</i>	In gradient space a smooth peak can be a regional minimum if its apex has less local variation than its slopes
<i>Marker-free watershed is reliable for clinical use</i>	Gilani et al. report catastrophic failure rates of 20–33% even under clean imaging conditions

Term	Definition
<i>Regional minimum</i>	Connected flat region with no lower adjacent pixel
<i>Catchment basin $CB(m_i)$</i>	All pixels whose steepest-descent path leads to m_i
<i>Watershed line $W(f)$</i>	Complement of all catchment basins; the boundary set
<i>Morphological gradient</i>	$g(f) = \text{dilation} - \text{erosion}$; measures local intensity spread
<i>Structuring element B</i>	Neighbourhood shape used in dilation/erosion
<i>Marker</i>	Connected pixel set seeding a catchment basin
<i>Geodesic reconstruction</i>	Morphological fill that eliminates unmarked minima
<i>Over-segmentation</i>	Too many basins due to spurious gradient minima
<i>Catastrophic failure</i>	Dice score < 0.1; complete segmentation breakdown
<i>WSHED</i>	Label assigned to a pixel whose neighbours belong to two or more distinct catchment basins; marks a watershed boundary point

References:

Beucher, S. (n.d.). Watershed, Hierarchical Segmentation and Waterfall Algorithm. Centre de Morphologie Mathématique, Mines Paris. Available at: https://cmm.minesparis.psl.eu/wp-content/uploads/2021/11/CMM_Morphonet_Beucher_watershed_hierarchical.pdf

The foundational paper establishing watershed segmentation within the mathematical morphology framework. Beucher formalises the topographic metaphor, greyscale image as terrain, pixel intensity as elevation, and defines regional minima, catchment basins, and watershed lines with mathematical precision. The flooding simulation is introduced as both a conceptual device and an algorithmic basis. The paper also presents the waterfall algorithm, a hierarchical extension that produces multi-scale segmentations by iteratively merging the shallowest watershed lines. Essential reading for understanding the theoretical foundations of all watershed-based methods.

Beucher, S. & Meyer, F. (1993). The Morphological Approach to Segmentation: The Watershed Transformation. In E. R. Dougherty (Ed.), *Mathematical Morphology in Image Processing* (pp. 433–481). Marcel Dekker. <https://www.taylorfrancis.com/chapters/edit/10.1201/9781482277234-12>

This chapter establishes the gradient-based watershed as the practical standard, arguing that the transform should be applied to the morphological gradient $g(f) = (f \oplus B) - (f \ominus B)$ rather than to the raw intensity image. The authors define dilation and erosion formally, prove the gradient's key properties, and demonstrate the approach on bubble segmentation. The priority-queue flooding algorithm is presented and its $O(n)$ complexity established. This work is the direct source of the morphological gradient formulation and flooding algorithm used throughout modern implementations.

Bai, M. & Urtasun, R. (2017). Deep Watershed Transform for Instance Segmentation. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5221–5229. <https://arxiv.org/abs/1611.08303>

Proposes replacing the hand-crafted morphological gradient with a learned energy map produced by a convolutional neural network, trained to predict direction-aware distance transforms that peak at object centres. The watershed cut is then applied to this learned surface rather than the image gradient, dramatically reducing over-segmentation on cluttered scenes. Evaluated on the CVPR instance segmentation benchmark, the method outperforms classical watershed and competitive proposal-based approaches. The paper validates the watershed framework as a sound conceptual foundation while demonstrating that its main limitation, sensitivity to the gradient surface, can be overcome with learned representations.

Gilani, Z. F., Mansoor, I., & Zhao, H. (2025). Reliability Auditing of Medical Image Segmentation under Signal Variability: The Catastrophic Failure Rate Framework. SSRN Preprint. <https://ssrn.com/abstract=5090029>

Introduces the Catastrophic Failure Rate (CFR) metric, defined as the proportion of test cases where Dice overlap falls below 0.1, as a reliability measure for segmentation methods under real-world signal degradation. Applied to marker-free watershed on dermoscopic images, the study reports CFRs of 20–33% under clean conditions, rising sharply under noise, blur, and contrast perturbation. The paper identifies boundary leakage at low-contrast edges as the dominant failure mode and argues that reliability auditing should be a standard component of any clinical deployment evaluation, alongside mean accuracy metrics.

Kaur, M. & Jindal, G. (2011). Medical Image Segmentation using Marker Controlled Watershed Transformation. *International Journal of Computer Science and Technology*, 2(4), 548–551. <https://www.ijcst.com/vol2/issue4/mandeep.pdf>

Presents a marker-controlled watershed pipeline applied to medical image segmentation, with particular emphasis on lung CT data. The paper introduces a five-step procedure: computing the morphological gradient, extracting foreground markers via opening-by-reconstruction, computing background markers from the distance-transformed binary image, imposing both marker sets on the gradient via geodesic reconstruction to eliminate spurious minima, and applying the watershed transform to the modified gradient. On benchmark lung CT slices, the method cleanly delineates anatomical structures with no spurious interior splits, in contrast to plain watershed which produces a heavily fragmented result.

Peng, Y. & Liu, R. (2010). Object Segmentation Based on Watershed and Graph Cut. *2010 3rd International Congress on Image and Signal Processing (CISP2010)*, pp. 1431–1435. <https://ieeexplore.ieee.org/document/5648034>

Proposes a hybrid segmentation method that uses watershed pre-segmentation to reduce the image into compact regions, then applies graph cut optimisation over those regions rather than individual pixels. The graph cut energy function $E = \sum V_{p,q} + \sum D_p$ encodes both boundary penalties and per-pixel data costs and is minimised globally using a max flow/min cut algorithm. On a 256×256 image, the hybrid approach reduces running time from 2.44 seconds (pixel-level graph cut) to 0.63 seconds while producing more accurate object boundaries than either watershed or graph cut alone, by combining watershed's speed with graph cut's global optimality.

Baswaraj, D., Govardhan, A. & Premchand, P. (2012). Active Contours and Image Segmentation: The Current State of the Art. *Global Journal of Computer Science and Technology*, 12(11). <https://computerresearch.org/index.php/computer/article/view/73>

Provides a comprehensive survey of active contour methods for image segmentation, covering both classical snake models, which deform an explicit set of contour points toward a local energy minimum, and level set methods, which represent the contour implicitly as the zero level of a higher-dimensional function and can handle topological changes such as splitting and merging. The paper distinguishes edge-based active contours, which rely on the image gradient and therefore inherit watershed's sensitivity to noise and local minima, from region-based active contours, which minimise global intensity statistics and require no specific contour initialisation. A key finding is that active contours always produce continuous boundary curves, whereas gradient-based edge detectors frequently produce discontinuous ones.